



YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING



Department of Computer Technology

Progress Seminar

Advanced monitoring and control mechanism for electric vehicle charging station using Machine learning

PRESENTED BY: Group CT- 23

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COLLEGE GUIDE:

Prof.N.U.Sambhe

SECTION : A & B

YEAR: 2024-25

Co-guide from Electrical Dept.

Prof.P.S.Shete

Department Vision and Mission

Vision

To be a well-known center for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary arena by developing problem-solving skills through emerging technologies.

Program Outcomes/Program Specific Outcomes

At the end of Program, Graduate students will have

- PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

Program Outcomes/Program Specific Outcomes

- PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
- PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
- PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

Program Outcomes/Program Specific Outcomes

- PO11: Lifelong Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies
- [PSO 1] An ability to acquire skills to design & develop quality software using cutting-edge technologies of Computer Engineering.
- [PSO 2] An ability to use knowledge of various domains as per thrust areas to formulate and implement ideas for providing innovative solutions.

Course Outcomes

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO 2
Identify a problem, analyze research gaps through a literature survey, and propose an ethical, innovative solution with societal impact.	3	3	3	3	3	2	2	3	3		2	3	2

Course Outcomes	Program Outcomes											Program Specific Outcomes	
	1	2	3	4	5	6	7	8	9	10	11	PSO 1	PSO 2
Develop a structured project report detailing the problem, methodology, proposed solution, and findings, adhering to professional and ethical standards.	3	3	3	3	3	3	2	3	3		2	3	3
Implement a basic prototype/model to validate the proposed solution, demonstrating practical application of knowledge and technical skills.	3	3			3			3	3	3	3	3	3
Avg	3	3	3	3	3	2.5	2	3	3	3	2.3	3	2.6667

Contents

1. Problem Definition
2. Introduction
3. Objectives
4. Work Done
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6. Status of Research Paper /Patent (If Any)
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Problem Statement

1. **Predicting Charger Current Output:** The need for a machine learning model to predict the optimal current output at EV charging stations, ensuring efficient charging.
2. **Fault Detection and Correction:** Addressing the challenge of real-time fault detection in charging equipment, with an automated control mechanism to correct faults or alert authorized personnel if correction is not possible.
3. **Optimizing Charging Operations:** Enhancing charging station performance by dynamically optimizing charging parameters to prevent load imbalances, reduce downtime, and ensure a seamless and efficient user experience.

Introduction

As electric vehicles (EVs) become an integral part of our daily lives, the need for smarter and more efficient charging infrastructure has never been greater. Charging stations must do more than just provide power — they need to be reliable, adaptable, and capable of meeting growing demand while minimizing energy waste.

Our project tackles this challenge by creating a predictive model using a data-driven approach and supervised machine learning. By analyzing both historical and real-time data, the model can forecast charging demand at public stations, allowing energy to be distributed efficiently, reducing grid stress, and enhancing the user experience.

But it doesn't stop there. The project also integrates advanced monitoring and control mechanisms, enabling real-time fault detection and better energy management. These features ensure charging stations remain efficient, safe, and capable of adapting to fluctuating energy loads and user needs.

Objectives

Primary Objective/Problem Definition:

To design and develop a data-driven predictive model using supervised machine learning regression methods to accurately forecast charging demand at public charging stations, and to propose advanced monitoring and control mechanisms to optimize energy management, reduce peak demand, and enhance the overall charging experience.

Specific Objectives:

1.To Collect Dataset: Gather historical data on electric vehicle charging patterns, energy consumption, and other relevant factors to support predictive modeling.

2.To design and develop model for Predictive analysis:Develop a machine learning-based predictive model that utilizes collected data to forecast charging demand and energy consumption.

3.To monitor and Predict: Design a real-time monitoring system that utilizes predictive analytics to forecast charging demand, detect anomalies, and optimize charging station operations.

4.To detect and manage fault: Develop a fault detection and management system that uses automated control mechanism to identify potential faults, alert operators, and provide recommendations for maintenance and repair.

Work Done

1. Data Acquisition

- Collected relevant dataset for the project
- Ensured data quality, completeness, and alignment with project goals

2. Data Preprocessing & Cleaning

- Removed duplicates, null values, and outliers
- Handled missing entries and standardized data formats
- Prepared structured data for analysis and visualization

3. Exploratory Data Analysis (EDA)

- Performed statistical analysis and summary of key parameters
- Visualized patterns and trends using multiple plots and graphs

4. **Graphical Analysis & Correlation Study**

- Plotted graphs to analyze relationships of:
 - Voltage vs. Time
 - Current vs. Time
 - Power vs. Time
 - Battery Temperature vs. Time
- Observed behavior and interaction of variables over charging session

5. **Initial Inferences & Observations**

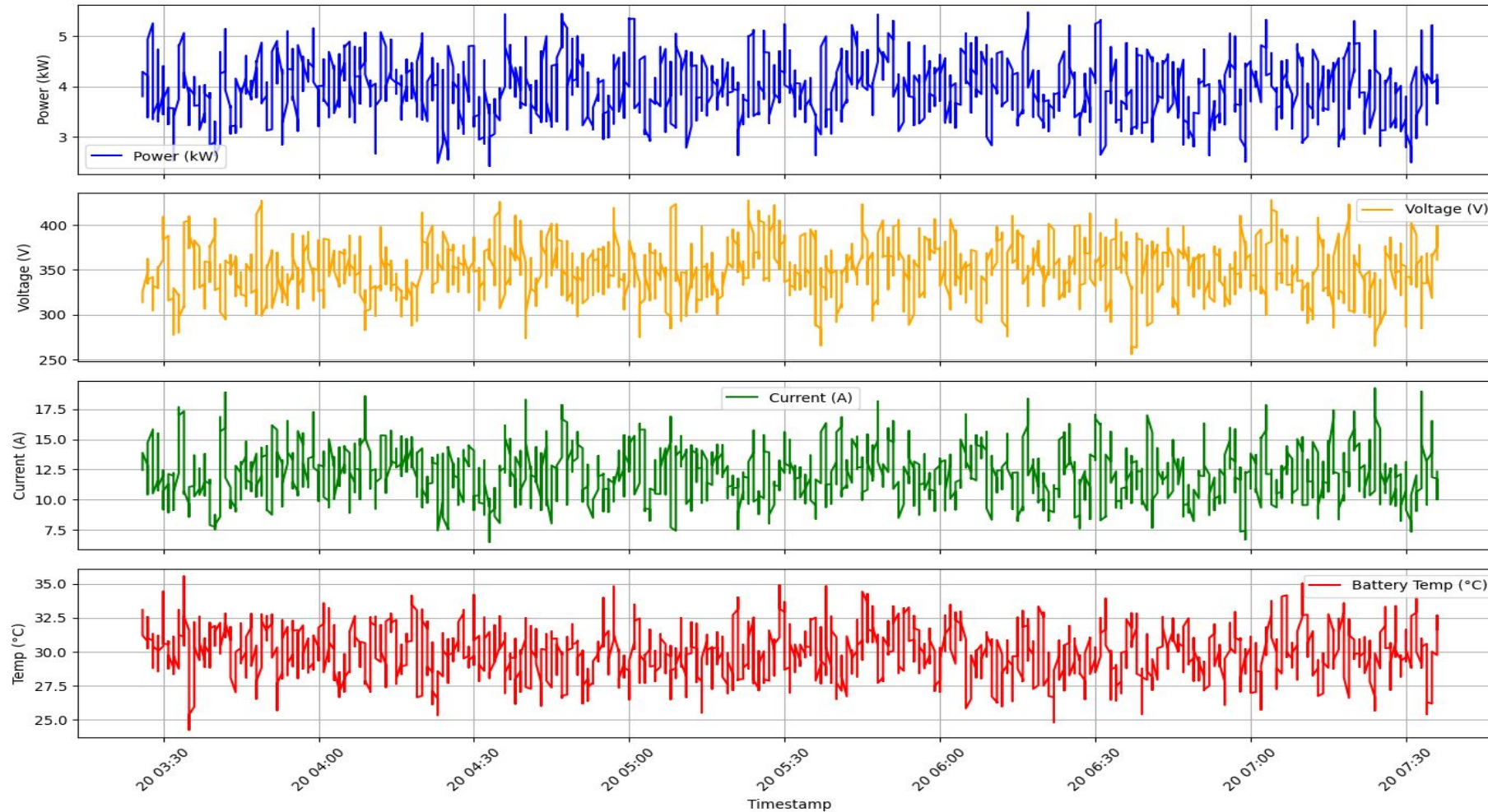
- Identified potential trends and anomalies in system behavior
- Noted key factors affecting charging efficiency and performance

Work Done

Inferences draw from the give data.

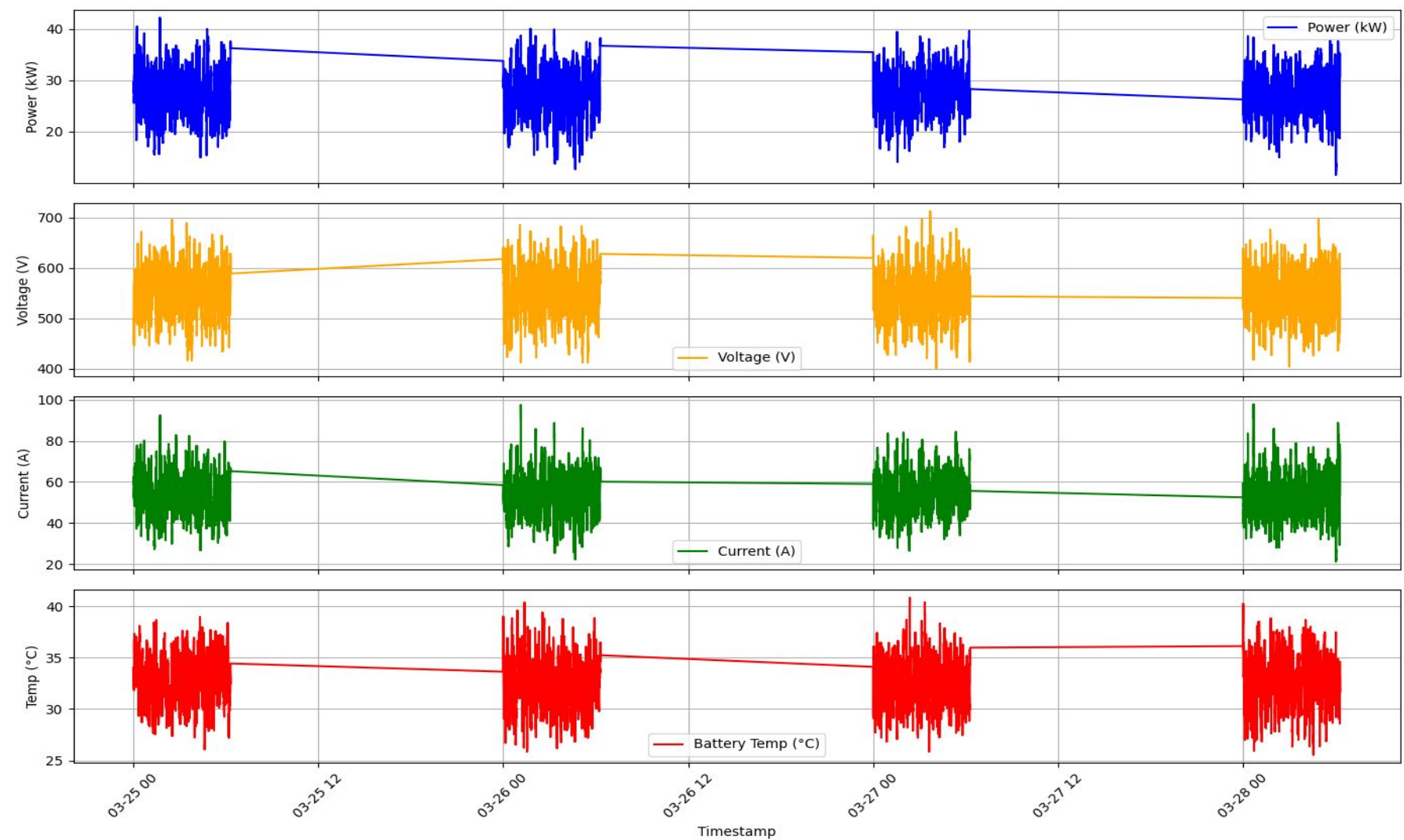
Graphs shows the relations of parameters with each others

Smoothed EV Charging Parameters Over Time



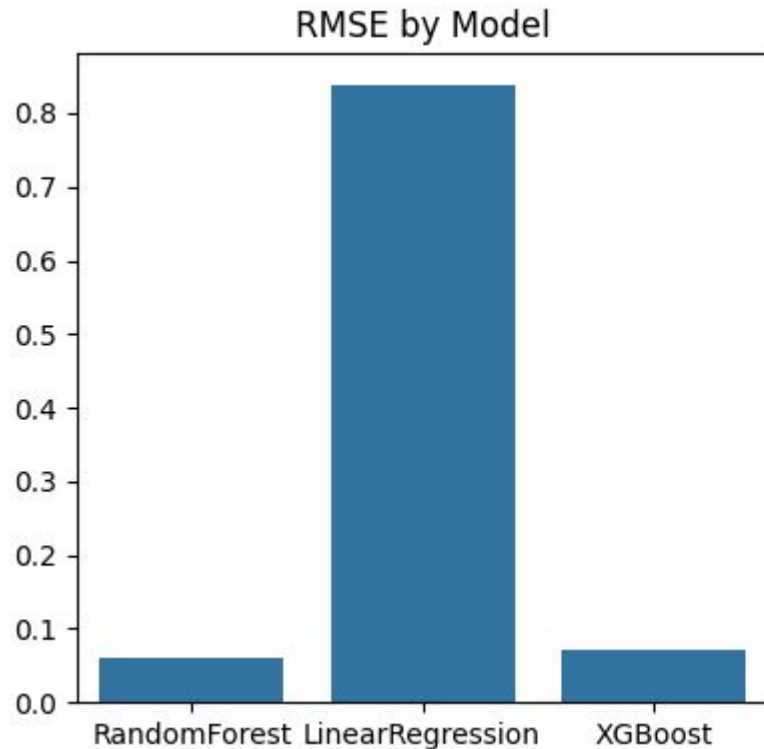
Work Done

Smoothed EV Charging Parameters Over Time

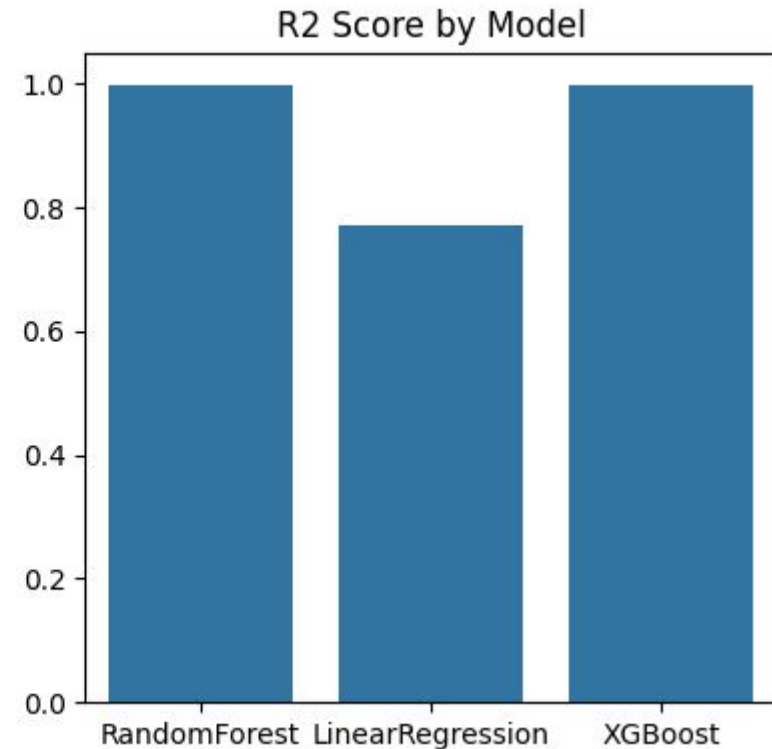


Work done

Analysis of different ML algorithm and their accuracy

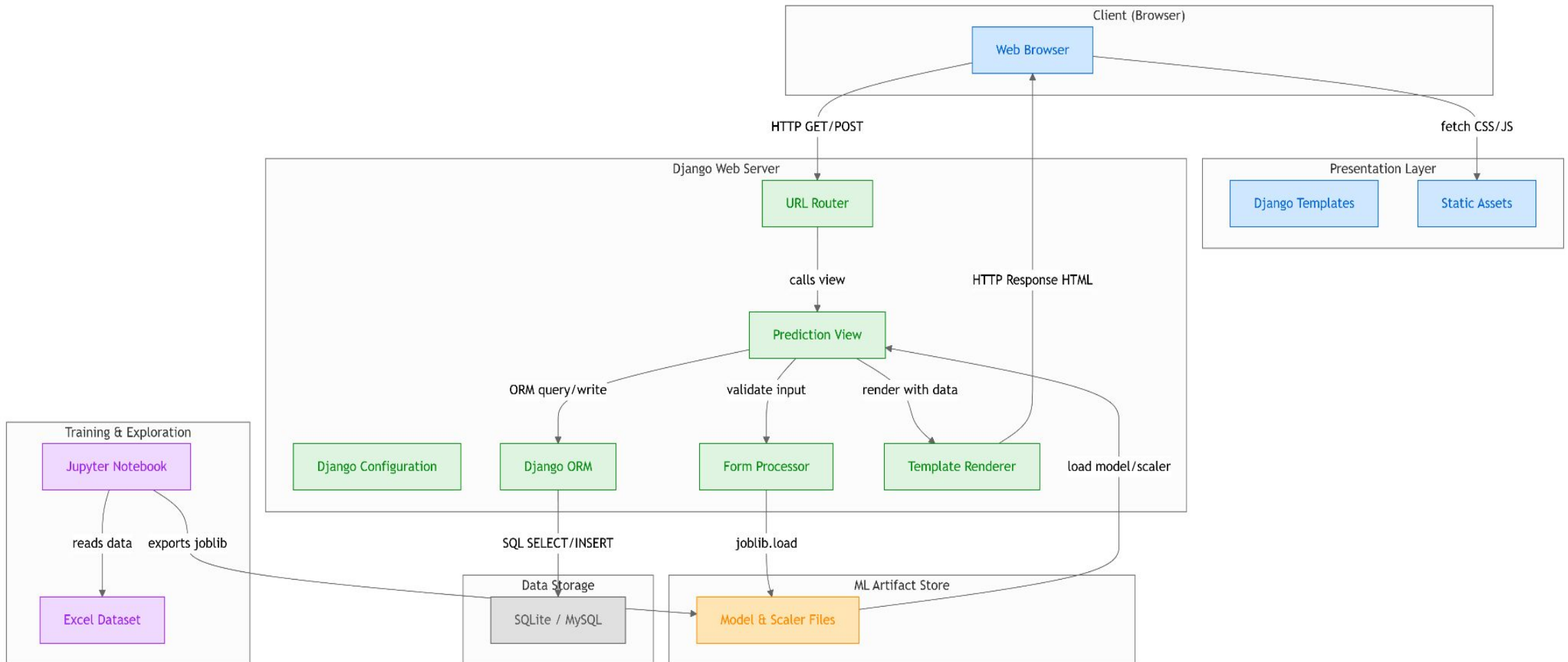


It gives the average prediction error in the original units of the output variable.



R^2 shows the proportion of the variance in the target variable that is explained by the model.

Work done



Expected Input and Output

← ↻ ⓘ 127.0.0.1:8000 🔍 ☆ ⋮

EV Charging Predictor

Main DC Power Distribution

EV 1EV 2EV 3

Total Power: 100 kW | Available: 0.0 kW

EV Charging Station 1

Charging Power (kW)*
50

Battery Temperature (°C)*
30

State of Charge (%)*
30

Charging Duration (hours)*
1

Charging Time*
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Prediction Results:
Power Allocated: 50.0 kW
Battery Temperature: 30.0°C
State of Charge: 30.0%
Duration: 1.0 hours
Energy Consumption: 4.42 kWh

EV Charging Station 2

Charging Power (kW)*
30

Battery Temperature (°C)*
35

State of Charge (%)*
50

Charging Duration (hours)*
2

Charging Time*
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Prediction Results:
Power Allocated: 30.0 kW
Battery Temperature: 35.0°C
State of Charge: 50.0%
Duration: 2.0 hours
Energy Consumption: 10.59 kWh

EV Charging Station 3

Charging Power (kW)*
20

Battery Temperature (°C)*
40

State of Charge (%)*
70

Charging Duration (hours)*
3

Charging Time*
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Prediction Results:
Power Allocated: 20.0 kW
Battery Temperature: 40.0°C
State of Charge: 70.0%
Duration: 3.0 hours
Energy Consumption: 13.30 kWh

Calculate Predictions

Status of Research Paper /Patent (If Any)

FORM 2 THE PATENT ACT 1970 (39 of 1970) & The Patents Rules, 2003 PROVISIONAL/COMPLETE SPECIFICATION (See section 10 and rule13)	
1. TITLE OF THE INVENTION: Advanced Monitoring and Control Mechanism for Electric Vehicle Charging Station using Machine Learning	
2. APPLICANT (S) (a) NAME: <u>Prof. N. U. Sambhe</u> (b) NATIONALITY: <u>Indian</u> (c) ADDRESS: <u>YCCE Nagpur</u>	
3. PREAMBLE TO THE DESCRIPTION	
PROVISIONAL EV charger uses machine learning to optimize power distribution and monitor faults in EV charging stations. It ensures safe operation by dynamically managing load and controlling charger connectivity.	COMPLETE This invention relates to the field of electric vehicle (EV) infrastructure, more specifically to a smart system that optimizes the monitoring and control of EV charging stations using machine learning algorithms.
4. DESCRIPTION <u>EVOptima</u> is an intelligent EV charging station system that uses machine learning to monitor and optimize power distribution across multiple charging points. It predicts electrical parameters, detects faults, and dynamically manages transformer load to ensure safety and efficiency.	
5. CLAIMS A machine learning-based system for optimizing power distribution in EV charging stations, adjusting current in real-time to prevent overloads. The system detects faults and automatically disconnects faulty chargers to maintain efficient operation.	
6. DATE AND SIGNATURE	

Advanced Monitoring and Control Mechanism for Electric Vehicle Charging Station Using Machine Learning

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Abstract - The increasing adoption of electric vehicles (EVs) worldwide has created an urgent need for scalable, intelligent, and energy-efficient charging infrastructure. Traditional charging mechanisms lack the adaptive capabilities necessary to manage unpredictable loads, fault tolerance, and optimal energy distribution. This paper introduces a novel framework combining real-time monitoring, fault detection, predictive analysis, and adaptive control using machine learning (ML) techniques to address these challenges. By employing supervised ML algorithms, we develop models that anticipate energy demand, optimize charging schedules, and enhance system safety. The proposed system architecture leverages cloud platforms, IoT sensors, and automated decision-making algorithms, aiming to deliver a user-

energy efficiency, operational reliability, and service quality at EV charging stations.

Index Terms— Electric Vehicles, Smart Charging, Machine Learning, Energy Management, Fault Detection, Predictive Analytics, Internet of Things (IoT), Smart Grids, Cloud Integration.

1 Introduction

Electric vehicles (EVs) represent a transformative shift in modern transportation, driven by the urgent need to reduce greenhouse gas emissions and dependence on fossil fuels. With the global EV market witnessing exponential growth, charging infrastructure is expected to scale accordingly. However, traditional EV charging stations often suffer from

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Thank You